Aerodry FBD-120 Fluid Bed Dryer

Operation and Maintenance Manual

Synthetic sample for PlantMemory

1.2 Safety and lockout

Fine powder in the dryer can ignite. Keep the machine earthed and never open the filter housing during a run.

1. Stop the drying cycle and wait until the exhaust fan has stopped.
2. Switch the main isolator QS1 to OFF, lock it and attach a danger tag.
3. Close the steam valve to the air heater and the air valve to the bowl seal.
4. Wait until the product temperature display reads below 40 °C.
5. Check that the earth clamp is fixed to the product bowl before any work.
6. Wear a dust mask and gloves as listed in the batch record.

2.1 Machine overview

Inlet air is heated and passes through the air distribution plate into the product bowl, where it fluidises the wet granules. The expansion chamber slows the air so particles fall back. Fines are held by the filter bags in the filter housing; a shaker bar shakes the bags at a set interval. The exhaust fan pulls the air through.

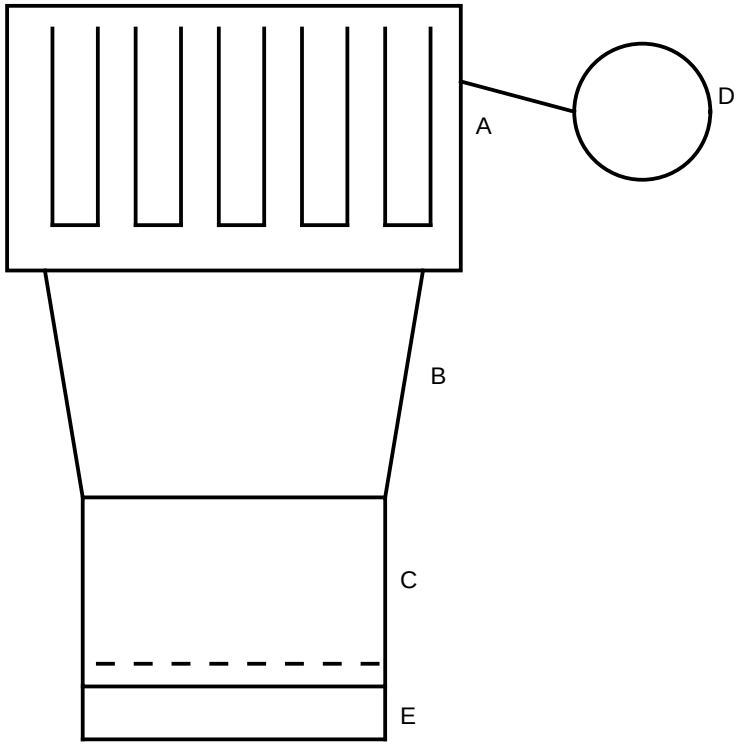

Fig. 2.1 A filter housing with bags, B expansion chamber, C product bowl, D exhaust fan, E inlet air.

4.1 Alarm list

Alarms starting with E stop the cycle. Alarms starting with W are warnings.

Code	Alarm	Effect	Typical repair
E-15	Filter differential pressure high	Stops	60 min
E-22	Inlet air temperature high	Stops	30 min
E-31	Earth (grounding) fault	Stops	15 min
E-40	Bowl seal pressure low	Stops	45 min
W-08	Filter differential pressure rising	Warning	-
W-12	Product temperature deviation	Warning	-

5.1 E-15 Filter differential pressure high

Symptom: alarm E-15, the pressure drop across the filter bags is above 25 mbar and the cycle stops. Likely causes: bags blinded by fines, shaking interval too long, a torn bag (fines in the exhaust duct), wet granules.

1. Lock out the dryer (section 1.2).
2. Open the filter housing and run the shaker manually for 2 minutes.
3. Inspect every bag for tears and holes, especially near the clamp ring.
4. Replace torn or blinded bags with filter bag set FBD-FB-120; fit the clamp evenly.
5. Check the exhaust duct for fines; if found, report to QA before restarting.
6. Set the shaking interval as per the product's batch record.

Typical time: 60 minutes.

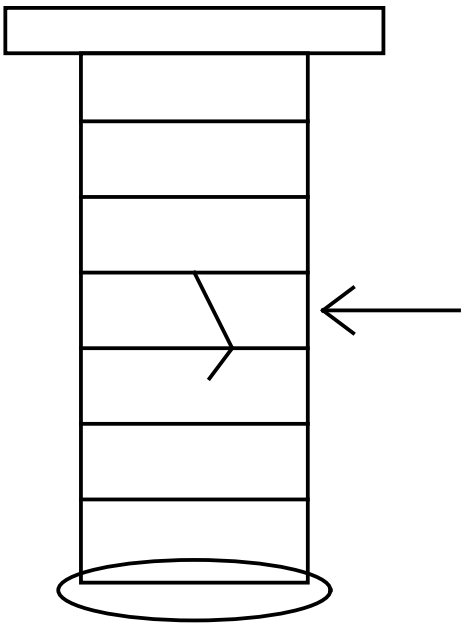

Fig. 5.1 Filter bag on the shaker bar. Check for tears near the clamp ring.

5.2 E-40 Bowl seal pressure low

Symptom: alarm E-40, the inflatable seal between bowl and expansion chamber does not reach 3 bar. Likely causes: a cut seal, low compressed air, a leaking air line.

1. Lock out the dryer (section 1.2) and lower the bowl.
2. Inspect the inflatable seal for cuts and flat spots.
3. Replace a damaged seal with bowl gasket FBD-GS-04.
4. Check compressed air supply is at least 5 bar.

Typical time: 45 minutes.

6.1 Return to service

1. Close the filter housing and remove locks and tags.
2. Run the fan empty for 5 minutes and check filter pressure drop is below 10 mbar.
3. QA checks the first granule sample (loss on drying) before the batch continues.
4. Record the repair and downtime in the equipment logbook.